Intelligent Load Scheduling

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1 Introduction

Demand response is an important approach for improving the efficiency of electricity consumption in residential areas. By scheduling household appliances according to electricity prices, it is possible to reduce energy costs and contribute to a more balanced load on the power grid. In this assignment, different demand response strategies are analyzed using optimization methods under Time-of-Use (ToU) and Real-Time Pricing (RTP) schemes.

To model realistic appliance scheduling, assumptions are made about the daily routine of the household. The availability of the user determines when certain appliances can reasonably be operated. The assumed daily schedule of the household is shown below:

- Awake at home (morning): 06:00–07:00
- Away from home: 07:00–17:00
- Home and awake: 17:00–23:00
- Sleeping: 23:00–06:00

These time windows are used to define realistic operating periods for shiftable appliances throughout the assignment. Appliances that require user interaction are scheduled during periods when the user is at home, while appliances that can operate automatically may run during nighttime hours.

2 Method

To answer all the questions, only a singular code was used. The questions differ mainly by their input appliances. Therefore, each task has an assosiated excel file containing the data of all relevant appliances. By inserting the correct question number, the code collects the relevant files and makes other necessary adjustments regarding electricity prices and L as required by the task.

Please find a detailed explanation of the codes in the "README" file provided in the ZIP file.

The code will create an optimal distribution for energy usage over all the hours in a day. That would normally entail a period from midnight to midnight. appliances such as the EV, is likely optimal to charge overnight. On the other hand, there are no appliances that are particularly useful to run while away from home. Therefore the code runs on a period from 10:00 am to 10:00 am. To simplify this with the collection of electricity prices the code therefore runs a simulation for a two-day period with all constraints set within the given period.

To make the solution better adjusted to the real pricing scheme we changed to resolution to 15 minute periods as was implemented in Norway in November of 2025. The original questions ask for an hourly resolution. Therefore, random prices and ToU prices are given with all quarters in the same hour at the same price. Within our model this is mathematically equivalent to an hourly resolution.

Furthermore, we added a function that collects the current electricity prices directly from NordPools (NP) websites. This has been used in addition to the prices given by random generation to provide a more functional application for load sceduling. Due to time constraints, the code simply repeats the current prices gathered for the current day to the entire two-day simulation period. A better code would collect the prices for the following day in the period from midnight to 10:00 am. Since the day-ahead prices are released at around 02:00 pm this would only work if the code was run after this point, and was therefore not implemented.

2.1 Appliance Constraints and Model Formulation

The appliance scheduling problem is formulated as an optimization problem where electricity consumption is distributed across a set of discrete time intervals. The objective of the model is to determine a feasible schedule for all appliances that minimizes the total electricity cost.

Let $x_{a,h}$ denote the electricity consumption of appliance a during time slot h. The optimization model determines the optimal values of $x_{a,h}$ for all appliances and time slots.

Each appliance must satisfy three main types of constraints.

2.1.1 Power level constraint

Each appliance has a minimum and maximum power level that can be used during a time slot. This constraint ensures that the scheduled power usage remains within the physical limits of the device:

$$x_a^{min} \leq x_{a,h} \leq x_a^{max}$$

where x_a^{min} and x_a^{max} represent the minimum and maximum power level of appliance a.

Such power constraints are commonly used in demand response models to represent appliance operating limits (6, p. 573).

2.1.2 Operating time constraint

Each appliance may only operate within a predefined time window. These windows represent practical limitations based on the user's daily routine or appliance functionality.

Let α_a denote the earliest start time and β_a the latest completion time for appliance a. The appliance may therefore only operate within the interval

$$\alpha_a \leq h \leq \beta_a$$

This constraint ensures that appliances are scheduled at realistic times and reflects typical assumptions used in home energy management systems (HEMS) (6, p. 573–574).

2.1.3 Total energy requirement

Each appliance requires a predefined total amount of energy during the scheduling horizon. Let E_a denote the total required energy consumption of appliance a. The model therefore enforces the constraint

$$\sum_h x_{a,h} = E_a$$

This ensures that the appliance receives enough energy to complete its operation while allowing the optimizer to determine when the energy is consumed.

Such energy balance constraints are standard in demand response scheduling models (7, p. 323–324).

3 Question 1

3.1 Introduction

Question 1 considers a simple household demand response problem with three shiftable appliances: a washing machine, a dishwasher and an electric vehicle (EV). The household is exposed to a Time-of-Use (ToU) pricing scheme, where electricity is more expensive during peak hours and cheaper during off-peak hours. In this case, the peak period is defined as 17:00–20:00 (5pm–8pm), while all remaining hours are treated as off-peak.

The purpose of the task is to determine a scheduling strategy that minimizes the total daily electricity cost while still satisfying the operating requirements of the appliances. This reflects the central idea of demand response, where consumers adjust their electricity usage in response to price signals in order to reduce costs and shift demand away from peak periods (6, p. 570–571).

3.2 Method

The scheduling problem is formulated as an optimization model where the objective is to minimize the total electricity cost over a daily scheduling horizon.

The household includes three appliances:

- Washing machine (1.94 kWh)
- Dishwasher (1.44 kWh)
- Electric vehicle (9.9 kWh)

Each appliance is assigned a feasible operating window based on typical household behaviour.

3.2.1 User availability and operating windows

To model realistic appliance scheduling, assumptions are made about the daily routine of the household. The availability of the user determines when certain appliances can reasonably be operated. The assumed daily schedule is:

- Awake at home (morning): 06:00–07:00
- Away from home: 07:00–17:00
- Home and awake (evening): 17:00–23:00
- Sleeping: 23:00–06:00

These time periods reflect a typical working day where the user leaves home around 07:00, works from 08:00 to 16:00, and returns home around 17:00 after commuting.

The availability of the user affects when different appliances can operate. For example, the washing machine must finish before sleeping so that the clothes can be handled afterwards. The dishwasher however, can run automatically during the night since it does not require user interaction once it has started.

Electric vehicles are typically equipped with smart charging systems that schedule charging automatically when electricity is available. In this model, EV charging is therefore restricted to periods when the user is at home rather than when the vehicle is parked at work.

Based on these assumptions, the following operating windows are used in the optimization model:

- Washing machine: 17:00–23:00
- Electric vehicle charging: 17:00–07:00
- Dishwasher: 19:00–01:00 and 06:00–09:00

The electricity price follows a Time-of-Use (ToU) pricing scheme. Electricity costs 1 NOK/kWh during the peak period between 17:00 and 20:00, while all other hours are priced at 0.5 NOK/kWh.

The optimization model determines the schedule that minimizes the total electricity cost while satisfying the operating constraints of the appliances.

3.3 Results

A snippet of the optimal scheduling result for Question 1 is shown in Figure 1.

```
Kl 21:00 - 21:15:   [washing_machine: 0.2kW]
Kl 21:15 - 21:30:   [washing_machine: 0.2kW]
Kl 21:30 - 21:45:   [washing_machine: 0.2kW]
Kl 21:45 - 22:00:   [washing_machine: 0.2kW]
Kl 22:00 - 22:15:   [washing_machine: 0.2kW]
Kl 22:15 - 22:30:   [washing_machine: 0.2kW]
Kl 22:30 - 22:45:   [washing_machine: 0.2kW]
Kl 22:45 - 23:00:   [washing_machine: 0.2kW]
Kl 23:00 - 23:15:
Kl 23:15 - 23:30:   [dishwasher: 0.1kW]
Kl 23:30 - 23:45:   [dishwasher: 0.1kW]
Kl 23:45 - 00:00:   [dishwasher: 0.1kW]
Kl 00:00 - 00:15:   [dishwasher: 0.1kW]
Kl 00:15 - 00:30:   [dishwasher: 0.1kW]
Kl 00:30 - 00:45:   [dishwasher: 0.1kW]
Kl 00:45 - 01:00:   [dishwasher: 0.1kW]
Kl 01:00 - 01:15:   [electric_vehicle: 0.9kW] [dishwasher: 0.1kW]
Kl 01:15 - 01:30:   [electric_vehicle: 0.9kW] [dishwasher: 0.1kW]
Kl 01:30 - 01:45:   [electric_vehicle: 0.9kW] [dishwasher: 0.1kW]
Kl 01:45 - 02:00:   [electric_vehicle: 0.9kW] [dishwasher: 0.1kW]
Kl 02:00 - 02:15:   [electric_vehicle: 0.9kW] [dishwasher: 0.1kW]
Kl 02:15 - 02:30:   [electric_vehicle: 0.9kW]
Kl 02:30 - 02:45:   [electric_vehicle: 0.9kW]
Kl 02:45 - 03:00:   [electric_vehicle: 0.9kW]
Kl 03:00 - 03:15:   [electric_vehicle: 0.9kW]
Kl 03:15 - 03:30:   [electric_vehicle: 0.9kW]
Kl 03:30 - 03:45:   [electric_vehicle: 0.9kW]
```

Figure 1: Snippet of the optimal appliance scheduling for Question 1.

The model produced a total electricity cost of 6.64 NOK.

3.3.1 Theoretical minimum cost

The total daily energy consumption of the three appliances is

$$E_{\text{total}} = 1.94 + 1.44 + 9.9 = 13.28 \text{ kWh}.$$

If all appliances are operated entirely during off-peak hours (0.5 NOK/kWh), the theoretical minimum cost becomes

$$C_{\text{theoretical}} = 13.28 \times 0.5 = 6.64 \text{ NOK}.$$

The optimization model achieves exactly this cost, meaning that all appliance usage is scheduled outside the peak period.

3.4 Discussion

The result shows that the optimal strategy is to schedule all appliances outside the peak pricing period. Since the electricity price during off-peak hours is half of the peak price, shifting consumption away from peak hours significantly reduces the total electricity cost.

The optimization model schedules the electric vehicle charging overnight, while the washing machine and dishwasher are placed in evening or nighttime hours. This completely avoids the peak period between 17:00 and 20:00.

Although the theoretical minimum cost is achieved, such a strategy may not always be practical in real households. Users may prefer to run appliances earlier in the evening rather than late at night. Nevertheless, the example illustrates how demand response can reduce electricity costs by shifting flexible electricity consumption in time.

4 Question 2

4.1 Introduction

In Question 2 the demand response model is extended from the simplified scenario in Question 1 to a more realistic household setting. Instead of only three appliances the household now contains a larger set of electrical devices such as lighting, heating, kitchen appliances, consumer electronics and charging devices.

The household is exposed to a Real-Time Pricing (RTP) electricity scheme. In contrast to Time-of-Use pricing where electricity prices are fixed within predefined peak and off-peak blocks, RTP reflects the dynamic nature of electricity markets where prices vary throughout the day depending on supply and demand conditions (6, p. 572–573). This type of pricing is commonly used in modern electricity markets and incentivizes consumers to shift flexible consumption to periods with lower prices.

The goal of the task is therefore to determine an optimal scheduling strategy for all household appliances that minimizes the total electricity cost while satisfying the operational constraints of each appliance.

4.2 Method

The optimization problem is formulated as a cost minimization problem over a scheduling horizon consisting of 15-minute time intervals. The use of smaller time intervals allows the model to represent electricity consumption more precisely and capture short-term price fluctuations.

The electricity price follows a randomly generated price curve which mimics the behavior of real electricity markets. The generated curve includes typical daily price variations with lower prices during nighttime hours and higher prices during periods with higher demand.

Figure 2: An overview of the pricing curves for NP73 and RND prices.

Each appliance is described by a set of parameters that define its operational constraints:

- Total daily energy requirement
- Duration of operation
- Allowed time windows
- Whether the appliance can be paused during operation
- Whether the power consumption can be adjusted

These constraints reflect realistic appliance behavior. For example, appliances such as washing machines and dryers must run continuously once started, while other appliances such as EV charging or heating can adjust their consumption over time.

Such appliance constraints are commonly used in demand response models and represent different categories of household loads, including shiftable appliances and must-run appliances (6, p. 574).

The optimization problem can therefore be written as

$$\min \sum_{t \in T} p_t \cdot l_t$$

where

- p_t is the electricity price in time slot t
- l_t is the total electricity consumption in time slot t

This formulation corresponds to a standard cost-minimization formulation used in home energy management systems (HEMS), where electricity consumption is scheduled automatically in response to time-varying electricity prices (6, p. 570–571).

4.3 Results

The optimization model provided a total cost of 37.9760 NOK for real prices of 07.03.2026. This price set will hereafter be referred to as NP73. A cut-out of the distribution is provided in figure 3.

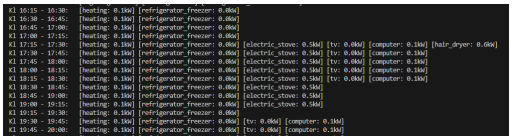


Figure 3: An overview of the load distribution for the conditions of question 2 with real prices.

For random prices the total cost was given by 23.2852 NOK. The set of random prices will hereafter be referred to as RND. A cut-out of the distribution is provided in figure 4.

Please note that the use of 4 decimal points is deliberate as it is needed to compare for task 4.

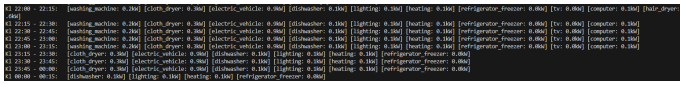


Figure 4: An overview of the load distribution for the conditions of question 2 with random prices.

4.4 Discussion

For both set of prices we observe the same characteristics as in question 1. None of the appliances have constraints affecting each other. Therefore, pausable appliances find the best set of time slots, while non-pausable find the best set of consecutive time slots. The model still lacks a way to properly account for spreading out the load, but this will be adressed in question 4.

When considering the results it is important to note that the price difference between the NP73 and RND prices is of no significans. Their creation is fundamentally different and can therefore not be subject to direct comparison. Anyhow, relative differences between the results of these between questions are interesting, and will be discussed in section 7.1.

5 Question 3

5.1 Introduction

In Question 3 the demand response model is extended from a single household to a neighbourhood consisting of 30 households. Each household has the same appliance configuration and operational constraints as in Question 2. However, only a fraction of the households are assumed to own an electric vehicle (EV).

The purpose of this task is to study how demand response scheduling behaves at a larger scale. Instead of optimizing the electricity consumption of a single household, the model now schedules the consumption of multiple households simultaneously.

The neighbourhood is exposed to a Real-Time Pricing (RTP) scheme, where electricity prices vary throughout the day. These prices are generated using a random price function that mimics realistic daily price fluctuations. The goal is therefore to determine an optimal scheduling strategy that minimizes the total electricity cost of the entire neighbourhood.

5.2 Method

The scheduling problem is formulated as an optimization problem where electricity consumption is distributed across a 24-hour period. Each household contains several appliances with predefined operational constraints such as energy requirements, duration of operation and allowed time windows.

Demand response models typically aim to shift electricity consumption in response to time-varying prices. Through price signals, consumers are incentivized to reschedule their energy usage patterns in order to reduce operating costs and peak demand (6, p. 570–571).

However, the model used in this assignment only considers the price signal from the electricity market to the consumers. The feedback from consumers to the market is therefore not modeled explicitly and the optimization only represents the demand-side response to given electricity prices.

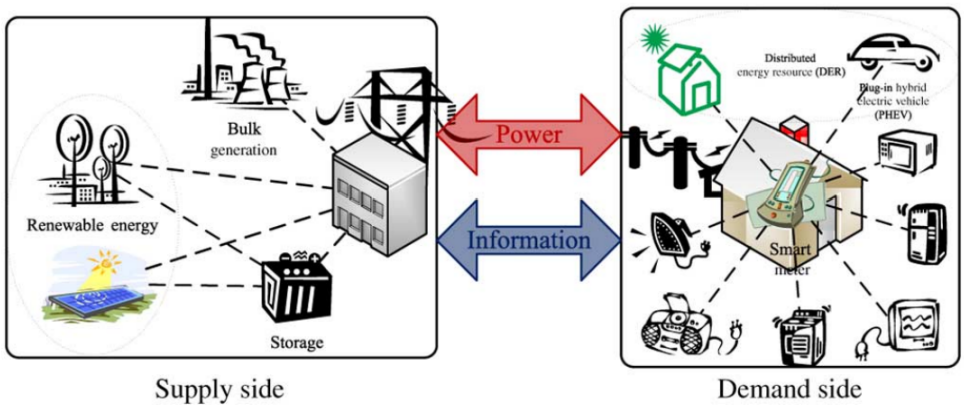


Figure 5: Conceptual illustration of two-way interaction between supply and demand in demand response systems (6, p. 570–571). (6, p. 570-571).

The optimization model minimizes the total electricity cost of the neighbourhood. Since electricity prices are given, the objective becomes minimizing the

cost of the aggregated electricity consumption:

$$\min \sum_{h \in H} p_h L_h$$

where p_h is the electricity price in time slot h and L_h is the total load of the neighbourhood at that time.

The total neighbourhood load is the sum of the loads from all households and appliances:

$$L_h = \sum_{n=1}^{30} \sum_{a \in A} x_{n,a}^h$$

where $x_{n,a}^h$ represents the electricity consumption of appliance a in household n at time slot h.

Each appliance must satisfy its operational constraints, including:

- Total energy requirement
- Power limits per time slot
- Allowed time windows

Such appliance-level constraints are typical in home energy management system (HEMS) models used for demand response scheduling (6, p. 573–574). Similar formulations are also used in distributed energy scheduling models for smart grids (7, p. 323–324).

5.3 Results

The optimization model provided a total cost of 1039.6715 NOK for real prices of 07.03.2026. For random prices the total cost was given by 644.9407 NOK. No cut-outs are provided in this question as the addition of 30 households make the cutouts difficult to read.

5.4 Discussion

The results of this question is practically identical to that of question 2. The only difference is the introduction of 29 additional households, some without an EV.

Because there is not introduced any limitations regarding peak capacity, and no appliances have constraints depending on other appliances, the same optimal set of timeslots will be identical between the households. Question 4 will introduce such a maximum capacity.

6 Question 4

6.1 Introduction

In Question 2 the optimization model was used to minimize the total electricity cost of a household under a Real-Time Pricing (RTP) scheme. A natural consequence of this type of scheduling is that several appliances may be shifted to the same low-price periods. While this can reduce the electricity bill, it may also create high power demand in certain time slots.

This issue is addressed in Question 4. The same household setting and the same price curve from Question 2 are used, but an additional upper bound is imposed on the total load in each time slot. In this way the scheduling model is forced not only to consider electricity cost, but also to avoid excessive peak consumption.

6.2 Method

In Question 4 the optimization model from Question 2 is extended with a peak-load constraint. The purpose is to limit the maximum electricity consumption within each time slot.

This modification is motivated by capacity-based grid tariffs, where consumers may be charged according to their highest power usage within a billing period. Such pricing models create incentives to avoid large consumption peaks and instead distribute electricity use more evenly over time.

To represent this behaviour in the optimization model, a peak limit L is introduced. Let ℓ_h denote the total electricity consumption in time slot h. The following constraint is imposed:

$$\ell_h \leq L \quad \forall h \in H$$

This ensures that the total electricity consumption in any time slot cannot exceed the peak limit.

The objective of the model remains the same as in Question 2, namely to minimize the total electricity cost:

$$\min \sum_{h \in H} p_h \ell_h$$

subject to the appliance constraints described earlier and the additional peak-load restriction.

This way the model may replicate the considerations of the Norwegian power grid fee (PGF). The current PGF model was introduced in 2022 and tracks the highest capacity used for each consumer in a single hour. The consumer is then placed into a bracket, and charged a one time fee at the end of the month according to the bracket. Since the model uses 15-minute time intervals the peak power limit (kW) can be translated into an energy limit per time slot.

The model is therefore solved for three different values of L:

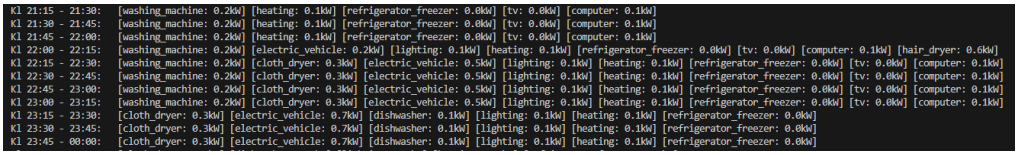
- $L = 5$ kW
- $L = 10$ kW
- $L = 15$ kW

These values correspond to our personal PGF. Since the model uses 15-minute time slots, the corresponding energy limits per slot become $L/4$.

6.3 Results

The optimization model provided a total cost of 37.9760 NOK for real prices of 07.03.2026 with a capacity of L=15kW and L=10kW. For a capacity of L=5kW the optimal cost is 37.9781 NOK.

The optimization model provided a total cost of 23.2852 NOK for random prices with a capacity of L=15kW and L=10kW. For a capacity of L=5kW the optimal cost is 23.5238 NOK. A cut-out of the distribution is provided in figure 6 and 7.



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K1 21:30 - 21:45:   [washing_machine: 0.2kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
K1 21:45 - 22:00:   [washing_machine: 0.2kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
K1 22:00 - 22:15:   [washing_machine: 0.2kW] [electric_vehicle: 0.2kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW] [hair_dryer: 0.6kW]
K1 22:15 - 22:30:   [washing_machine: 0.2kW] [cloth_dryer: 0.3kW] [electric_vehicle: 0.5kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
K1 22:30 - 22:45:   [washing_machine: 0.2kW] [cloth_dryer: 0.3kW] [electric_vehicle: 0.5kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
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K1 23:00 - 23:15:   [washing_machine: 0.2kW] [cloth_dryer: 0.3kW] [electric_vehicle: 0.5kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
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K1 23:45 - 00:00:   [cloth_dryer: 0.3kW] [electric_vehicle: 0.7kW] [dishwasher: 0.1kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW]
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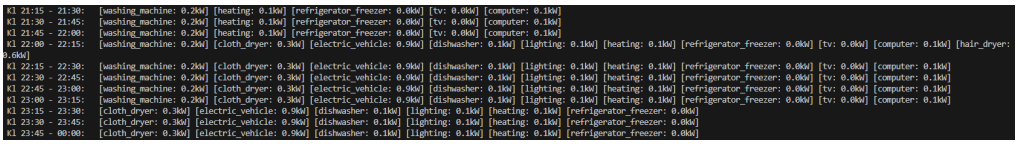
Figure 6: An overview of the load distribution for the conditions of question 4 with random prices and capacity L=5kW.



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K1 21:30 - 21:45:   [washing_machine: 0.2kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
K1 21:45 - 22:00:   [washing_machine: 0.2kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
K1 22:00 - 22:15:   [washing_machine: 0.2kW] [cloth_dryer: 0.3kW] [electric_vehicle: 0.9kW] [dishwasher: 0.1kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW] [hair_dryer:
0.6kW]
K1 22:15 - 22:30:   [washing_machine: 0.2kW] [cloth_dryer: 0.3kW] [electric_vehicle: 0.9kW] [dishwasher: 0.1kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
K1 22:30 - 22:45:   [washing_machine: 0.2kW] [cloth_dryer: 0.3kW] [electric_vehicle: 0.9kW] [dishwasher: 0.1kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
K1 22:45 - 23:00:   [washing_machine: 0.2kW] [cloth_dryer: 0.3kW] [electric_vehicle: 0.9kW] [dishwasher: 0.1kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
K1 23:00 - 23:15:   [washing_machine: 0.2kW] [cloth_dryer: 0.3kW] [electric_vehicle: 0.9kW] [dishwasher: 0.1kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW] [tv: 0.0kW] [computer: 0.1kW]
K1 23:15 - 23:30:   [cloth_dryer: 0.3kW] [electric_vehicle: 0.9kW] [dishwasher: 0.1kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW]
K1 23:30 - 23:45:   [cloth_dryer: 0.3kW] [electric_vehicle: 0.9kW] [dishwasher: 0.1kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW]
K1 23:45 - 00:00:   [cloth_dryer: 0.3kW] [electric_vehicle: 0.9kW] [dishwasher: 0.1kW] [lighting: 0.1kW] [heating: 0.1kW] [refrigerator_freezer: 0.0kW]
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Figure 7: An overview of the load distribution for the conditions of question 4 with random prices and capacity L=10kW.

6.4 Discussion

The first major takeaway for the results is the relative difference in the results between the L=5 and L=10 solutions for the hourly and quarterly resolutions. As mentioned earlier, the prices themselves are not comparable, but the relative differences are.

In both cases the L=10 solutions are equivalent to the optimal solutions from question 2. Limiting the maximum usage to 5kW only increased the total cost

by around 0.002 and 0.2 NOK or 0.005% and 1% respectively. This result is evidence of the profound impact that switching to a 15min resolution has. With this increased resolution there are far more cheap quarters throughout the day that can be used to find near optimal solution without utilizing a large capacity in any given time-slot.

To find the real optimal solution for whether or not to move up a PGF bracket, one would need to compare the price of moving up a bracket with the cost savings of utilizing the cheaper time slots over the entire month. Since future electricity prices for the entire month are unknown, the decision of whether to move to a higher capacity bracket must be made under uncertainty. This effectively turns the problem into a non-deterministic decision problem.

The problem could theoretically find a solution by evaluating a large set of potential prices to find the probability of cutting cost over the entire month by moving up a bracket. However, this is likely not a very rewarding endeavor. Looking at the results it is unlikely that there will ever be an optimal solution that prefers moving up an PGF bracket. The price of which is in the hundreds of NOK for the month, while the expected benefit is very low.

It is important to note that this may change. Our results are only given for a singular day, and can therefor not be used as strong evidence in itself. With the rising occurrence of potential negative prices and increasing price volatility, the picture painted by our results may be different.

7 Analysis of the impact of ToU and RTP pricing schemes on energy cost

Two different electricity pricing schemes are considered in this report: Time-of-Use (ToU) pricing and Real-Time Pricing (RTP). These pricing models influence how flexible electricity consumption can be scheduled and therefore affect the achievable energy cost.

Time-of-Use pricing divides the day into predefined price periods such as off-peak and peak hours. Electricity prices remain constant within each period but differ between them. Consumers can therefore reduce their electricity costs by shifting flexible consumption from expensive peak periods to cheaper off-peak periods.

Real-Time Pricing (RTP), on the other hand, reflects the dynamic nature of electricity markets where prices vary throughout the day depending on supply and demand conditions (6, p. 572–573). This pricing scheme provides more detailed price signals and allows consumers to schedule electricity consumption more precisely according to market conditions.

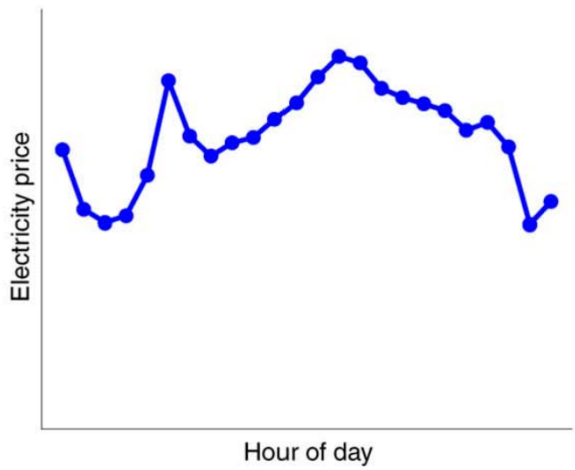

Figure 8: Example of a Real-Time Pricing (RTP) electricity price curve (6, p. 572).

In the experiments conducted in this report, the ToU pricing scheme is represented by the simplified peak and off-peak prices used in Question 1. The RTP scheme is represented by the generated price curves used in Questions 2–4. These curves mimic realistic daily price fluctuations while allowing the

optimization model to react to more detailed price signals.

In general, RTP creates a more complex optimization problem than ToU because the model must determine the optimal set of time slots under continuously varying prices. However, it also provides greater flexibility for shifting loads to low-price periods and can therefore lead to more efficient scheduling of flexible electricity consumption.

7.1 Overall discussion

An overview of all the results is provided in table 1.

Question	**Real**	**RND**
1		6,64
2	37,98	23,29
3	1039,67	944,94
4 (5)	37,9781	23,5238
4 (10)	37,9760	23,2852
4 (15)	37,9760	23,2852

Table 1: Overview of all results.

The main findings of the report can be summarized as follows:

- RTP makes for a more complex optimization problem than ToU as it must find an optimal set of timeslots to produce within.
- The introduction of more actors into the optimization problem does not entail any major difference in our experiments, as it did not include capacity limits or real 2 way price signals.
- The NP73 and RND prices show major differences in optimization effectiveness given capacity limits due to their different resolution. Our results gave a near optimal solution only 0.005% more expensive for a 15min resolution, while the hourly resolution yielded a 1% price increase.

References

[1] Office of Energy Efficiency, Natural Resources Canada. *Energy Consumption of Household Appliances Shipped in Canada*. December 2005. Available at: https://oee.nrcan.gc.ca/Publications/statistics/cama05/pdf/cama05.pdf

[2] Energy Use Calculator. *Household Appliance Energy Consumption Calculator*. Available at: http://energyusecalculator.com/

[3] Energy Use Calculator. *Electricity Usage of a Wi-Fi Router*. Available at: https://energyusecalculator.com/electricity_wifirouter.htm

[4] Energy Use Calculator. *Electricity Usage of a Coffee Maker*. Available at: https://energyusecalculator.com/electricity_coffeemaker.htm

[5] Energy Use Calculator. *Electricity Usage of a Cell Phone Charger*. Available at: https://energyusecalculator.com/electricity_cellphone.htm

[6] R. Deng, Z. Yang, M.-Y. Chow, and J. Chen, "A Survey on Demand Response in Smart Grids: Mathematical Models and Approaches," *IEEE Transactions on Industrial Informatics*, vol. 11, no. 3, pp. 570–582, 2015. Available at: https://www.researchgate.net/publication/276916545

[7] A.-H. Mohsenian-Rad, V. W. S. Wong, J. Jatskevich, R. Schober, and A. Leon-Garcia, "Autonomous Demand-Side Management Based on Game-Theoretic Energy Consumption Scheduling for the Future Smart Grid," *IEEE Transactions on Smart Grid*, vol. 1, no. 3, pp. 320–331, 2010. Available at: https://ieeexplore-ieee-org.ezproxy.uio.no/document/5628271

Appendix A Datasets from Excel

Prices from NordPool 07.03.2026:

00:00 - 00:	117,76
00:15 - 00:	117,39
00:30 - 00:	116,15
00:45 - 01:	115,25
01:00 - 01:	115,77
01:15 - 01:	114,71
01:30 - 01:	113,74
01:45 - 02:	112,96
02:00 - 02:	113,7
02:15 - 02:	113,16
02:30 - 02:	111,89
02:45 - 03:	111,46
03:00 - 03:	112,56
03:15 - 03:	111,79
03:30 - 03:	111,45
03:45 - 04:	111,21
04:00 - 04:	111,63
04:15 - 04:	111,77
04:30 - 04:	111,85
04:45 - 05:	112,57
05:00 - 05:	112,1
05:15 - 05:	113,58
05:30 - 05:	113,89
05:45 - 06:	114,46
06:00 - 06:	112,78
06:15 - 06:	112,99
06:30 - 06:	113,02
06:45 - 07:	112,97
07:00 - 07:	117,54
07:15 - 07:	115,39
07:30 - 07:	113,16
07:45 - 08:	109,72
08:00 - 08:	112,93
08:15 - 08:	110,86
08:30 - 08:	107,95
08:45 - 09:	106,15
09:00 - 09:	107,49
09:15 - 09:	106,04
09:30 - 09:	105,12
09:45 - 10:	103,92
10:00 - 10:	103,04
10:15 - 10:	101,78
10:30 - 10:	100,18
10:45 - 11:	97,33
11:00 - 11:	98,42
11:15 - 11:	98,79
11:30 - 11:	97,97

11:45 - 12:0	97,88
12:00 - 12:1	96,98
12:15 - 12:3	96,54
12:30 - 12:4	96,36
12:45 - 13:0	96,32
13:00 - 13:1	95,35
13:15 - 13:3	95,44
13:30 - 13:4	95,67
13:45 - 14:0	95,85
14:00 - 14:1	95,42
14:15 - 14:3	95,6
14:30 - 14:4	96,28
14:45 - 15:0	97,38
15:00 - 15:1	98,08
15:15 - 15:3	101,16
15:30 - 15:4	103,88
15:45 - 16:0	104,54
16:00 - 16:1	105,68
16:15 - 16:3	106,94
16:30 - 16:4	109,63
16:45 - 17:0	111,96
17:00 - 17:1	109,44
17:15 - 17:3	112,63
17:30 - 17:4	120,81
17:45 - 18:0	123,97
18:00 - 18:1	126,52
18:15 - 18:3	129,06
18:30 - 18:4	134,77
18:45 - 19:0	135,68
19:00 - 19:1	149,58
19:15 - 19:3	144,87
19:30 - 19:4	128,2
19:45 - 20:0	128,93
20:00 - 20:1	127,45
20:15 - 20:3	126,91
20:30 - 20:4	123,56
20:45 - 21:0	123,45
21:00 - 21:1	123,73
21:15 - 21:3	123,54
21:30 - 21:4	121,48
21:45 - 22:0	120,64
22:00 - 22:1	119,67
22:15 - 22:3	118,34
22:30 - 22:4	116,87
22:45 - 23:0	116,15
23:00 - 23:1	114,65
23:15 - 23:3	114,64

23:30 - 23:	114,3
23:45 - 00:	112,03
00:00 - 00:	117,76
00:15 - 00:	117,39
00:30 - 00:	116,15
00:45 - 01:	115,25
01:00 - 01:	115,77
01:15 - 01:	114,71
01:30 - 01:	113,74
01:45 - 02:	112,96
02:00 - 02:	113,7
02:15 - 02:	113,16
02:30 - 02:	111,89
02:45 - 03:	111,46
03:00 - 03:	112,56
03:15 - 03:	111,79
03:30 - 03:	111,45
03:45 - 04:	111,21
04:00 - 04:	111,63
04:15 - 04:	111,77
04:30 - 04:	111,85
04:45 - 05:	112,57
05:00 - 05:	112,1
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05:30 - 05:	113,89
05:45 - 06:	114,46
06:00 - 06:	112,78
06:15 - 06:	112,99
06:30 - 06:	113,02
06:45 - 07:	112,97
07:00 - 07:	117,54
07:15 - 07:	115,39
07:30 - 07:	113,16
07:45 - 08:	109,72
08:00 - 08:	112,93
08:15 - 08:	110,86
08:30 - 08:	107,95
08:45 - 09:	106,15
09:00 - 09:	107,49
09:15 - 09:	106,04
09:30 - 09:	105,12
09:45 - 10:	103,92
10:00 - 10:	103,04
10:15 - 10:	101,78
10:30 - 10:	100,18
10:45 - 11:	97,33
11:00 - 11:	98,42

11:15 - 11:	98,79
11:30 - 11:	97,97
11:45 - 12:	97,88
12:00 - 12:	96,98
12:15 - 12:	96,54
12:30 - 12:	96,36
12:45 - 13:	96,32
13:00 - 13:	95,35
13:15 - 13:	95,44
13:30 - 13:	95,67
13:45 - 14:	95,85
14:00 - 14:	95,42
14:15 - 14:	95,6
14:30 - 14:	96,28
14:45 - 15:	97,38
15:00 - 15:	98,08
15:15 - 15:	101,16
15:30 - 15:	103,88
15:45 - 16:	104,54
16:00 - 16:	105,68
16:15 - 16:	106,94
16:30 - 16:	109,63
16:45 - 17:	111,96
17:00 - 17:	109,44
17:15 - 17:	112,63
17:30 - 17:	120,81
17:45 - 18:	123,97
18:00 - 18:	126,52
18:15 - 18:	129,06
18:30 - 18:	134,77
18:45 - 19:	135,68
19:00 - 19:	149,58
19:15 - 19:	144,87
19:30 - 19:	128,2
19:45 - 20:	128,93
20:00 - 20:	127,45
20:15 - 20:	126,91
20:30 - 20:	123,56
20:45 - 21:	123,45
21:00 - 21:	123,73
21:15 - 21:	123,54
21:30 - 21:	121,48
21:45 - 22:	120,64
22:00 - 22:	119,67
22:15 - 22:	118,34
22:30 - 22:	116,87
22:45 - 23:	116,15

23:00 - 23:	114,65
23:15 - 23:	114,64
23:30 - 23:	114,3
23:45 - 00:	112,03

Randomly generated prices:

00:00 - 00:15	67,71
00:15 - 00:30	67,71
00:30 - 00:45	67,71
00:45 - 01:00	67,71
01:00 - 01:15	62,82
01:15 - 01:30	62,82
01:30 - 01:45	62,82
01:45 - 02:00	62,82
02:00 - 02:15	65,51
02:15 - 02:30	65,51
02:30 - 02:45	65,51
02:45 - 03:00	65,51
03:00 - 03:15	66,38
03:15 - 03:30	66,38
03:30 - 03:45	66,38
03:45 - 04:00	66,38
04:00 - 04:15	69,15
04:15 - 04:30	69,15
04:30 - 04:45	69,15
04:45 - 05:00	69,15
05:00 - 05:15	83,26
05:15 - 05:30	83,26
05:30 - 05:45	83,26
05:45 - 06:00	83,26
06:00 - 06:15	88,09
06:15 - 06:30	88,09
06:30 - 06:45	88,09
06:45 - 07:00	88,09
07:00 - 07:15	88,48
07:15 - 07:30	88,48
07:30 - 07:45	88,48
07:45 - 08:00	88,48
08:00 - 08:15	87,69
08:15 - 08:30	87,69
08:30 - 08:45	87,69
08:45 - 09:00	87,69
09:00 - 09:15	94,5
09:15 - 09:30	94,5
09:30 - 09:45	94,5
09:45 - 10:00	94,5
10:00 - 10:15	81,28
10:15 - 10:30	81,28
10:30 - 10:45	81,28
10:45 - 11:00	81,28
11:00 - 11:15	90,02
11:15 - 11:30	90,02
11:30 - 11:45	90,02
11:45 - 12:00	90,02
12:00 - 12:15	93,38
12:15 - 12:30	93,38

12:30 - 12:45	93,38
12:45 - 13:00	93,38
13:00 - 13:15	81,22
13:15 - 13:30	81,22
13:30 - 13:45	81,22
13:45 - 14:00	81,22
14:00 - 14:15	83,05
14:15 - 14:30	83,05
14:30 - 14:45	83,05
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15:45 - 16:00	73,31
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16:45 - 17:00	80,39
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00:00 - 00:15	67,71
00:15 - 00:30	67,71
00:30 - 00:45	67,71
00:45 - 01:00	67,71

01:00 - 01:15	62,82
01:15 - 01:30	62,82
01:30 - 01:45	62,82
01:45 - 02:00	62,82
02:00 - 02:15	65,51
02:15 - 02:30	65,51
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03:15 - 03:30	66,38
03:30 - 03:45	66,38
03:45 - 04:00	66,38
04:00 - 04:15	69,15
04:15 - 04:30	69,15
04:30 - 04:45	69,15
04:45 - 05:00	69,15
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05:45 - 06:00	83,26
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10:45 - 11:00	81,28
11:00 - 11:15	90,02
11:15 - 11:30	90,02
11:30 - 11:45	90,02
11:45 - 12:00	90,02
12:00 - 12:15	93,38
12:15 - 12:30	93,38
12:30 - 12:45	93,38
12:45 - 13:00	93,38
13:00 - 13:15	81,22
13:15 - 13:30	81,22

13:30 - 13:45	81,22
13:45 - 14:00	81,22
14:00 - 14:15	83,05
14:15 - 14:30	83,05
14:30 - 14:45	83,05
14:45 - 15:00	83,05
15:00 - 15:15	73,31
15:15 - 15:30	73,31
15:30 - 15:45	73,31
15:45 - 16:00	73,31
16:00 - 16:15	80,39
16:15 - 16:30	80,39
16:30 - 16:45	80,39
16:45 - 17:00	80,39
17:00 - 17:15	92,49
17:15 - 17:30	92,49
17:30 - 17:45	92,49
17:45 - 18:00	92,49
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21:15 - 21:30	72,29
21:30 - 21:45	72,29
21:45 - 22:00	72,29
22:00 - 22:15	58,57
22:15 - 22:30	58,57
22:30 - 22:45	58,57
22:45 - 23:00	58,57
23:00 - 23:15	59,79
23:15 - 23:30	59,79
23:30 - 23:45	59,79
23:45 - 00:00	59,79